32. **a.** If $p \neq 2$, then $1 \neq p-1$ in $\mathbb{Z}_p$, but $1^2 = (p-1)^2$. Thus the squaring function mapping $\mathbb{Z}_p \to \mathbb{Z}_p$ is not one to one; in fact, its image can have at most $p-1$ elements. Thus some element of $\mathbb{Z}_p$ is not a square if $p \neq 2$.

b. We saw in Example 29.19 that there is a finite field of four elements. Let p be an odd prime. By Part(**a**), there exists $a \in \mathbb{Z}_p$ such that $x^2 - a$ has no zeros in $\mathbb{Z}_p$. This means that $x^2 - a$ is irreducible in $\mathbb{Z}_p[x]$. Let α be a zero of $x^2 - a$ in an extension field of $\mathbb{Z}_p$. By Exercise 30, $\mathbb{Z}_p(\alpha)$ has p^2 elements.

33. Let $\beta \in F(\alpha)$. Then β is equal to a quotient $r(\alpha)/s(\alpha)$ of polynomials in α with coefficients in F. Suppose that $f(\beta) = 0$ where $f(x) \in F[x]$ and is of degree n. Multiplying the equation $f(\beta) = 0$ by $s(\alpha)^n$, we obtain a polynomial in α with coefficients in F which is equal to zero. But then, α is algebraic over F, which is contrary to hypothesis. Therefore there is no such nonzero polynomial expression $f(\beta) = 0$, that is, β is transcendantal over F.

34. We know that $x^3 - 2$ is irreducible in $\mathbb{Q}[x]$ by the Eisenstein condition with $p = 2$. Therefore $\sqrt[3]{2}$ is algebraic of degree 3 over $\mathbb{Q}$. By Theorem 29.18, the field $\mathbb{Q}(\sqrt[3]{2})$ consists of all elements of $\mathbb{R}$ of the form $a + b(\sqrt[3]{2}) + c(\sqrt[3]{2})^2$ for $a, b, c \in \mathbb{Q}$, and distinct values of a, b, and c give distinct elements of $\mathbb{R}$. The set given in the problem consists of precisely these elements of $\mathbb{R}$, so the given set is the field $\mathbb{Q}(\sqrt[3]{2})$.

35. We keep using Theorem 29.18 and Exercise 30. Now the polynomial $x^3 + x + 1$ in $\mathbb{Z}_2[x]$ has no zeros in $\mathbb{Z}_2$ and is therefore irreducible in $\mathbb{Z}_2[x]$. If α is a zero of this polynomial in an extension field, then $\mathbb{Z}_2(\alpha)$ has $2^3 = 8$ elements by Exercise 30.

Similarly, let α be a zero of the irreducible polynomial $x^4 + x + 1$ in $\mathbb{Z}_2[x]$. Then $\mathbb{Z}_2(\alpha)$ has $2^4 = 16$ elements.

Finally, let α be a zero of the irreducible polynomial $x^2 - 2$ in $\mathbb{Z}_5[x]$. Then $\mathbb{Z}_5(\alpha)$ has $5^2 = 25$ elements.

36. Following the hint, we let F^* be the multiplicative group of nonzero elements of F. We are given that F is finite; suppose that F has m elements. Then F^* has $m-1$ elements. Because the order of an element of a finite group divides the order of the group, we see that for all $a \in F^*$ we have $a^{m-1} = 1$. Thus every $a \in F^*$ is a zero of the polynomial $x^{m-1} - 1$. Of course, 0 is a zero of x. Thus every $\alpha \in F$ is algebraic over the prime field $\mathbb{Z}_p$ of F, for the polynomial $x^{m-1} - 1$ is in $\mathbb{Z}_p$ for all primes p.

37. Let E be a finite field with prime subfield $\mathbb{Z}_p$. If $E = \mathbb{Z}_p$, then the order of E is p and we are done. Otherwise, let $\alpha_1 \in E$ where $\alpha_1 \notin \mathbb{Z}_p$. Let $F_1 = \mathbb{Z}_p(\alpha_1)$. By Exercise 30, the field F_1 has order p^{n_1} where n_1 is the degree of α_1 over $\mathbb{Z}_p$. If $F_1 = E$, we are done, Otherwise, we find $\alpha_2 \in E$ where $\alpha_2 \notin F_1$, and form $F_2 = F_1(\alpha_2)$, obtaining a field of order $p^{n_1 n_2}$ where n_2 is the degree of α_2 over F_1. We continue this process, constructing fields F_i of order $p^{n_1 n_2 \cdots n_i}$. Because E is a finite field, this process must eventually terminate with a field $F_r = E$. Then E has order $p^{n_1 n_2 \cdots n_r}$ which is a power of p as asserted.

30. Vector Spaces

1. $\{(0,1),(1,0)\}$, $\{(1,1),(-1,1)\}$, and $\{(2,1),(1,2)\}$

2. Suppose that $a(1,\,1,\,0) + b(1,\,0,\,1) + c(0,\,1,\,1) = (d, e, f)$. Then $a + b = d, a + c = e$, and $b + c = f$. Subtracting the second equation from the first, we obtain $b - c = d - e$. Adding this to the last equation, we obtain $2b = f + d - e$, so $b = (f + d - e)/2$. Then $a = (d + e - f)/2$ and $c = (e + f - d)/2$.

This shows that the given vectors span $\mathbb{R}^3$. Setting $d = e = f = 0$, we see that we must then have $a = b = c = 0$ so the vectors are also independent, and hence are a basis for $\mathbb{R}^3$.

3. We claim the vectors are dependent, and thus cannot form a basis. If a(-1, 1, 2) + b(2, -3, 1) + c(10, -14, 0) = (0, 0, 0), then

$$\begin{array}{rcrcrcl} -a & + & 2\text{b} & + & 10c & = & 0 \\ a & - & 3b & - & 14c & = & 0 \\ 2a & + & b & & & = & 0. \end{array}$$

Adding the first two equations, we find that $-b - 4c = 0$. Adding twice the first equation to the last, we find that $5b + 20c = 0$, which is essentially the same equation. Let $c = 1$ so $b = -4$ and $a = 2$. We find that

$$2(-1, 1, 2) + (-4)(2, -3, 1) + 1(10, -14, 0) = (0, 0, 0),$$

so the vectors are indeed independent.

4. Because $\sqrt{2}$ is a zero of irreducible $x^2 - 2$ of degree 2, Theorem 30.23 shows that a basis is $\{1, \sqrt{2}\}$.

5. Because $\sqrt{2}$ is in $\mathbb{R}$ and is a zero of $x - \sqrt{2}$ of degree 1, Theorem 30.23 shows that a basis is $\{1\}$.

6. Because $\sqrt[3]{2}$ is a zero of irreducible $x^3 - 2$ of degree 3, by Theorem 30.23 a basis is $\{1, \sqrt[3]{2}, (\sqrt[3]{2})^2\}$.

7. Because $\mathbb{C} = \mathbb{R}(i)$ where i is a zero of irreducible $x^2 + 1$ of degree 2, Theorem 30.23 shows that a basis is $\{1, i\}$.

8. Because i is a zero of irreducible $x^2 + 1$ of degree 2, Theorem 30.23 shows that a basis is $\{1, i\}$.

9. Since $\sqrt[4]{2}$ is a zero of irreducible $x^4 - 2$ of degree 4, by Theorem 30.23 a basis is $\{1, \sqrt[4]{2}, \sqrt{2}, (\sqrt[4]{2})^3\}$.

10. Recall that α is a zero of $x^2 + x + 1$, so $\mathbb{Z}_2(\alpha)$ is a 2-dimensional vector space over $\mathbb{Z}_2$. Thus the three elements $1, 1 + \alpha$, and $(1 + \alpha)^2 = 1 + 2 \cdot \alpha + \alpha^2 = 1 + 1 + \alpha = \alpha$ must be independent. By inspection, we see that

$$1(1) + 1(1 + \alpha) + 1(1 + \alpha)^2 = 1 + (1 + \alpha) + \alpha = 0,$$

so α is a zero of $x^2 + x + 1$, which is thus not only $\text{irr}(\alpha, \mathbb{Z}_2)$ but is also $\text{irr}(1 + \alpha, \mathbb{Z}_2)$. Of course, we already knew this because the polynomial's other zero, bedsides α, must lie in $\mathbb{Z}_2(\alpha)$ since $x^2 + x + 1$ has to factor into linear factors there, and $1 + \alpha$ is the only possibility for the other zero. However, we wanted to present the technique given in our first argument.

11. The definition is incorrect. Delete the "uniquely".

The vectors in a subset S of a vector space V over a field F **span** V if and only if each $\beta \in V$ an be expressed as a linear combination of the vectors in S.

12. The definition is incorrect. Some coefficients in the linear combination must be nonzero.

The vectors in a subset S of a vector space V over a field F are **linearly independent over** F if and only if the zero vector cannot be expressed as a linear combination, having some coefficients nonzero, of vectors in S.

13. The definition is correct.

14. The definition is incorrect. Replace "dependent" by "independent".

A **basis** for a vector space V over a field F is a set of vectors in V that span V and are linearly independent.

15. T F T T F F F T T T

16. a. A **subspace** of the vector space V over F is a subset W of V that is closed under vector addition and under multiplication by scalars in F, and is itself a vector space over F under these two operations.

b. Let $\{W_i \mid i \in I\}$ be a collection of subspaces of V. Because $\langle W_i, + \rangle$ is an abelian group, Theorem 7.4 shows that $\bigcap_{i\in I} W_i$ is again an abelian group. Let $a \in F$ and let $\alpha \in \bigcap_{i\in I} W_i$. Then $\alpha \in W_i$ for each $i \in I$, so $a\alpha \in W_i$ for each $i \in I$ because each W_i is a vector space. Hence $a\alpha \in \bigcap_{i\in I} W_i$ so the intersection is closed under scalar multiplication. All the other axioms for a vector space (distributive laws, etc.) certainly hold in this intersection, because they hold for all elements in V.

17. a. Let S be a subset of a vector space V over a field F. The **subspace generated by** S is the intersection of all subspaces of V that contain S.

b. Clearly, the sum of two finite linear combinations of elements of S is again a finite linear combination of elements of S. Also, a scalar times a finite linear combination is again a finite linear combination:

$$a(b_1\alpha_1 + \cdots + b_n\alpha_n) = (ab_1)\alpha_n + \cdots + (ab_n)\alpha_n.$$

Because $0 = 0\alpha$ for $\alpha \in S$, we see that $0 \in V$ is a finite linear combination of elements of S. Multiplying by the scalar -1, we see that an additive inverse of such a linear combination is again a finite linear combination of elements of S. Therefore the set of all finite linear combinations of elements of S is a vector space, and is clearly the smallest vector space that contains S.

This result is analogous to the case of Theorem 7.6 for abelian groups, although we do have to check here that scalar multiplication is closed.

18. The direct sum of vector spaces $V_1, V_2, \cdots, V_n$ over the same field F is

$$\{(\alpha_1, \alpha_2, \cdots, \alpha_n) \mid \alpha_1 \in V_i \text{ for } i = 1, 2, \cdots, n\},$$

with addition and scalar multiplication defined by

$$(\alpha_1, \alpha_2, \cdots, \alpha_n) + (\beta_1, \beta_2, \cdots, \beta_n) = (\alpha_1 + \beta_1, \alpha_2 + \beta_2, \cdots, \alpha_n + \beta_n).$$

and

$$a(\alpha_1, \alpha_2, \cdots, \alpha_n) = (a\alpha_1, a\alpha_2, \cdots, a\alpha_n) \text{ for all } a \in F.$$

Because addition and multiplication are defined by performing the operations in each component, and because the vectors appearing in each component form a vector space over F, it is clear that this direct sum is again a vector space.

19. Let F be any field and let $F^n = \{(a_1, a_2, \cdots, a_n) \mid a_i \in F\}$. Then F^n is a vector space with addition and multiplication of n-tuples defined by performing those operations in each component. (It is the direct sum of F with itself n times, as defined in Exercise 18.) A basis for F^n is

$$\{(1, 0, 0, \cdots, 0), (0, 1, 0, \cdots, 0), \cdots, (0, 0, 0, \cdots, 1)\}.$$

20. Let V and V' be vector spaces over the same field F. A map $\phi : V \to V'$ is an **isomorphism** if ϕ is a one-to-one map, $\phi[V] = V'$, and furthermore

$$\phi(\alpha + \beta) = \phi(\alpha) + \phi(\beta) \quad \text{and} \quad \phi(a\alpha) = a\phi(\alpha)$$

for all $\alpha, \beta \in V$ and all $a \in F$.

21. Because each vector in V can be expressed as a linear combination of the β_i, we see that $\{\beta_1, \beta_2, \cdots, \beta_n\}$ generates V. Now $0 = 0\beta_1 + 0\beta_2 + \cdots + 0\beta_n$. By hypothesis, this is the *unique* linear combination of the β_i that yields 0, so the vectors β_i are independent. Therefore, $\{\beta_1, \beta_2, \cdots, \beta_n\}$ is a basis for V.

For the other direction, suppose that $\{\beta_1, \beta_2, \cdots, \beta_n\}$ is a basis for V. Then every vector α is a linear combination of the β_i. Suppose that $\alpha = \sum_{i=1}^{n} c_i\beta_i$ and also $\alpha = \sum_{i=1}^{n} d_i\beta_i$. Subtracting, we obtain $0 = \alpha - \alpha = \sum_{i=1}^{n}(c_i - d_i)\beta_i$. Because the β_i are linearly independent, we must have $c_i - d_i = 0$ so $c_i = d_i$ for $i = 1, 2, \cdots n$, and the expression for α as a linear combination of the β_i is unique.

22. **a.** The system can be rewritten as

$$X_1\alpha_1 + X_2\alpha_2 + \cdots + X_n\alpha_n = \beta \tag{1}$$

because the ith component of the vector on the left side of the equation (1) is $a_{i1}X_1 + a_{i2}X_2 + \cdots + a_{in}X_n$ and the ith component of β is β_i. Equation (1) shows that the system has a solution if and only if β is a finite linear combination of the vectors α_j for $j = 1, 2, \cdots, n$. By Exercise 17, this means that the system has a solution if and only if β lies in the subspace generated by the vectors $\alpha_1, \alpha_2, \cdots, \alpha_n$.

b. Note that if $\{\alpha_j \mid j = 1, \cdots, n\}$ is a basis for a vector space, then Exercise 21 shows that the linear combination of the α_j that equals a vector β is unique. In terms of Part(**a**), this means that the system (1) has a unique solution.

23. Let $\{\gamma_1, \gamma_2, \cdots, \gamma_n\}$ be a basis for V. Let $\phi : F^n \to V$ be defined by $\phi(a_1, a_2, \cdots, a_n) = a_1\gamma_1 + a_2\gamma_2 + \cdots + a_n\gamma_n$. Because addition and multiplication in F^n is by components, it is obvious that $\phi(\alpha + \beta) = \phi(\alpha) + \phi(\beta)$ and $\phi(a\alpha) = a\phi(\alpha)$ for all $\alpha, \beta \in V$ and all $a \in F$. Because $\{\gamma_1, \gamma_2, \cdots, \gamma_n\}$ is a basis for V, every vector in V can be expressed as a linear combination of these vectors, so ϕ maps F^n *onto* V. By Exercise 21, the expression for a vector in V as a linear combination of the vectors γ_i is *unique*, so ϕ is one to one. Thus ϕ is an isomorphism.

24. **a.** Let $\alpha \in V, \alpha \neq 0$. Because $\{\beta_i \mid i \in I\}$ is a basis for V, we know by Exercise 21 that there are *unique* vectors $\beta_{i_1}, \beta_{i_2}, \cdots, \beta_{i_n}$ and nonzero scalars $a_1, a_2, \cdots a_n$ such that $\alpha = a_1\beta_{i_1} + a_2\beta_{i_2} + \cdots + a_n\beta_{i_n}$. By the conditions for a linear transformation, we then have

$$\phi(\alpha) = a_1\phi(\beta_{i_1}) + a_2\phi(\beta_{i_2}) + \cdots + a_n\phi(\beta_{i_n}).$$

This shows that the map ϕ is completely determined by the values $\phi(\beta_i)$ for $i \in I$.

b. Let $\phi : V \to V'$ be defined as follows: For nonzero $\alpha \in V$, express α as a linear combination

$$\alpha = a_1\beta_{i_1} + a_2\beta_{i_2} + \cdots + a_n\beta_{i_n} \tag{2}$$

with nonzero scalars $a_1, a_2, \cdots a_n$. This can be done because $\{\beta_i \mid i \in I\}$ is a basis for V. Define $\phi(\alpha) = a_1\beta_{i_1}{}' + a_2\beta_{i_2}{}' + \cdots + a_n\beta_{i_n}{}'$. Because the expression (2) for α with nonzero scalars is unique by Exercise 21, we see that ϕ is well defined, and of course, $\phi(\beta_i) = \beta_i{}'$ for $i \in I$. Because addition and scalar multiplication of linear combinations of the β_i and the $\beta_i{}'$ are both achieved by adding and scalar multiplying respectively the coefficients in the linear combinations, we see at once that ϕ satisfies the required properties for a linear transformation. Part(**a**) shows that this transformation is completely determined by the vectors $\beta_i{}'$, that is, the linear transformation is unique.